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Literature Survey

Helmut Simonis

Constraint Based Production Scheduling



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Key Points



- We are working on a survey of the existing CP & Scheduling literature
- Considers over 1200 papers
- Current version of survey available at <https://hsimonis.github.io/pthg24>



CP and Scheduling Literature Survey

- Methodology

- Analysis Results

- Limitations

Summary

A Survey of the Existing Literature



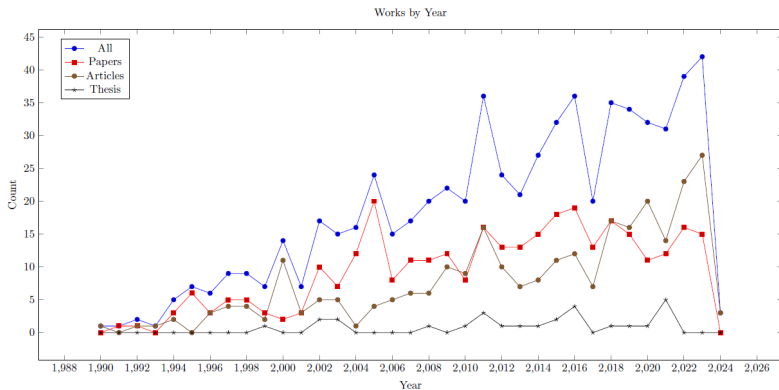
- Joint work with Cemalettin Ozturk, MTU
- What is out there
- Where to start
- Where to publish
- I'm interested in some specific topic, what is relevant

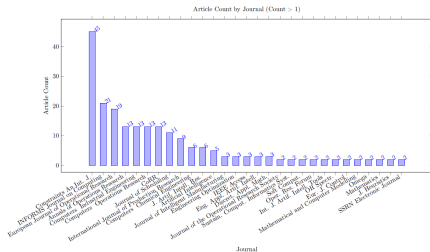


- Manually curated list of works, somewhat inclusive
- Starting with bibtex files
- Citation links through OpenCitations (open access)
- Content analysis on local copies of pdf files
- Closure of domain by analyzing missing cited and citing works
- Limited manual analysis of works (datasets, code)
- Results presented as LaTeX documents
- Open source analysis on git:

`https://hsimonis.github.io/pthg24/`

Overall Analysis (Based on 671 Works)





Most Recent Articles



Table 5: Works from bibtex (Total 274)

Key	Authors	Title	LC	Cite	Year	Conference / Journal	Pages	Nr Cites	Nr Refs	b	c
ForbesHJST24 ForbesHJST24	M. Forbes, M. Harris, H. Jansen, F.A. van der Schoot, T. Jansen	Combining optimisation and simulation using logic-based Benders decomposition	Yes	217	2024	European Journal of Operational Research	15	0	26	1314	1496
PrataAN23 PrataAN23	Bruno A. Prata, Levi R. Abreu, Marcelo S. Nagano	Applications of constraint programming in production scheduling problems: A descriptive bibliometric analysis	Yes	599	2024	Results in Control and Optimization	17	0	0	1427	1497
abs-2402-00459 abs-2402-00459	S. Nguyen, Dhananjay R. Thiruvady, Y. Sun, M. Zhang	Genetic-based Constraint Programming for Resource Constrained Job Scheduling	Yes	469	2024	CoRR	21	0	0	1495	1498
AbreuNP23 AbreuNP23	Levi Ribeiro de Abreu, Marcelo Seido Nagano, Bruno A. Prata	A new two-stage constraint programming approach for open shop scheduling problem with machine blocking	Yes	169	2023	International Journal of Production Research	20	1	47	1243	1499
AbreuPNF23 AbreuPNF23	Levi R. Abreu, Bruno A. Prata, Marcelo S. Nagano, Jose M. Framinan	A constraint programming-based iterated greedy algorithm for the open shop with sequence-dependent processing times and makespan minimization	Yes	3	2023	Computers Operations Research	12	0	46	1244	1500
Adelgren2023 Adelgren2023	N. Adelgren, Christos T. Maravelias	On the utility of production scheduling formulations including record keeping variables	Yes	71	2023	Computers Industrial Engineering	12	0	43	1245	1501
AtsarVPG23 AtsarVPG23	S. Afari, Camino R. Vela, Juan José Palacios, J. González-Rodríguez	Mathematical models and benchmarking for the fuzzy job shop scheduling problem	Yes	8	2023	Computers Industrial Engineering	14	0	50	1246	1502
AkramNHISA23 AkramNHISA23	Ulfat Omar Akram, Nor Kamariah Noordin, P. Hashim, Mohd Fadzli A. Rasid, Mustafa Ismail	Joint Scheduling and Routing Optimization for Deterministic Hybrid Traffic in Time-Sensitive Networks Using Constraint Programming	Yes	13	2023	IEEE Access	16	0	0	1248	1503
AlfieriGP23 AlfieriGP23	A. Alfieri, M. Garraffa, E. Pastore, F. Salassa	Permutation flowshop problems minimizing core waiting time and core idle time	Yes	19	2023	Computers Industrial Engineering	13	0	37	1249	1504
Cabalero23 Cabalero23	Jordi Coll Cabalero	Scheduling through logic-based tools	Yes	127	2023	Engineering Constraints An Int. J.	1	0	0	1287	1505
CierniowskaWZ23 CierniowskaWZ23	K. Cierniowska, R. Wichniarek, K. Zywicki	Constraint Programming for Flexible Flow Shop Scheduling Problem with Repeated Jobs and Repeated Operations	Yes	159	2023	Advances in Science and Technology Research Journal	14	0	0	1297	1506
FahimiQ23 FahimiQ23	H. Fahimi, C. Quimper	Overload-Checking and Edge-Finding for Robust Cumulative Scheduling	No	207	2023	INFORMS Journal on Computing	ml	0	16	No	1507
Fatemi-AnarakiTFV23 Fatemi-AnarakiTFV23	S. Fatemi-Anaraki, R. Tavakkoli-Moghaddam, M. Fourniani, B. Valodji-Nouri	Scheduling of Multi-Robot Job Shop Systems in Dynamic Environments: Mixed-Integer Linear Programming and Constraint Programming Approaches	Yes	212	2023	Omega	15	7	60	1312	1508
GhasemiMH23 GhasemiMH23	S. Ghasemi, R. Tavakkoli-Moghaddam, M. Hamid	Operating room scheduling by emphasizing human factors and dynamic decision-making styles: a constraint programming method	No	242	2023	International Journal of Systems Science: Operations Logistics	ml	0	104	No	1509
GuoZ23 GuoZ23	P. Guo, J. Zhu	Capacity reservation for humanitarian relief: A logic-based Benders decomposition method with subgradient cut	Yes	269	2023	European Journal of Operational Research	29	0	112	1325	1510
GurPAE23 GurPAE23	S. Güler, M. Pinarbasi, Hacı Mehmet Alakus, F. Eren	Operating room scheduling with surgical team: a new approach with constraint programming and goal programming	Yes	270	2023	Central Eur. J. Oper. Res.	25	1	40	1327	1511
IsikYA23 IsikYA23	Eyyup Ensar Isik, Soyda Topaloglu Yildiz, Öge	Constraint programming models for the hybrid flow shop scheduling problem and its extension	Yes	321	2023	Soft Comput.	28	0	127	1350	1512
JuvinIH.23a JuvinIH.23a	C. Juvin, L. Housini, P. Lopez	Logic-based Benders decomposition for the preemptive flexible job-shop scheduling problem	Yes	331	2023	Computers Operations Research	17	0	40	1355	1513
LacknerMMW23 LacknerMMW23	M. Lackner, C. Mikviclav, N. Musil, D. Walkowicz, F. Winter	Exact methods for the Oven Scheduling Problem	Yes	374	2023	Constraints An Int. J.	42	0	32	1371	1514

Automaticaly Extracted Article Features



Table 6: Automatically Extracted ARTICLE Properties (Requires Local Copy)

Work	Pages	Concepts	Classification	Constraints	Prog Languages	CP Systems	Areas	Industries	Benchmarks	Algorithm	a	c
Lactorio03 369	38	task, precedence, order, cmax, machine, job, activity, re-scheduling, setup-time, release-date, inventory, preempt, job-shop, resource, scheduling, make-span		cycle, table constraint, cumulative, disjunctive	C++	Ilog Scheduler			benchmark	edge-finding, not-last, energetic reasoning, not-first, time-tabling	1201	1731
LactorioRSV18 372	41	release-date, job-shop, resource, activity, precedence, sequence dependent setup, earliness, scheduling, machine, inventory, transportation, manpower, due-date, setup-time, batch process, order, tardiness, flow-shop, job, make-span, re-scheduling, task, distributed	psplib, parallel machine, RCPSP	alternative constraint, cumulative, noOverlap, disjunctive, span constraint, cycle, alwaysIn, endBeforeStart	C, Python, C++, Java	CHIP, Gecode, Ilog Solver, Cplex, Ilog Scheduler, OPL, Choco Solver, CPO	semiconductor railway, container terminal, satellite, robot, pipeline, aircraft, shipping line	chemical industry, petro-chemical industry	real-world, CSPLib, benchmark	edge-finding	1080	1610
LacknerMMWW23 374	42	release-date, batch process, setup-time, job, order, due-date, tardiness, scheduling, make-span, machine, task, lateness, job-shop, earliness	parallel machine, OSP, single machine	alternative constraint, disjunctive, bin-packing, noOverlap, cumulative, endBeforeStart		Chuffed, Cplex, OPL, CPO, OR-Tools, MiniZinc, Gurobi	oven scheduling	electronics industry, steel industry, manufacturing industry	random instance, industrial partner, benchmark, instance generator, zenodo, real-life	time-tabling	954	1514
LammaMM97 377	15	job-shop, resource, scheduling, precedence, order, task, job, distributed, no-wait		circuit, disjunctive	C++, Prolog	ECLiPSe, OPL, CHIP	railway				1230	1760
LatorCB15 385	52	machine, make-span, job, precedence, resource, scheduling, task, order	psplib	cumulative, cycle, bin-packing	Java, Prolog	Choco Solver, CHIP, SICStus, Ilog Solver, OZ, Cplex, ECLiPSe, OPL, CHIP			generated instance, Roadef, benchmark, random instance	energetic reasoning, sweep, edge-finding	1110	1640
LiW08 386	18	precedence, activity, resource, completion-time, setup-time, make-span, scheduling, machine, preempt, job-shop, no preempt, job, re-scheduling, open-shop, due-date, task, order	RCPSP	disjunctive, cycle, bin-packing					real-world		1178	1708
LieseM08 388	12	preempt, resource, scheduling, machine, job, activity, precedence, job-shop, task, make-span, order, cmax	RCPSP, psplib	disjunctive, cumulative	C++	OZ			benchmark	edge-finding	1179	1709
LimtanyakuS12 393	32	release-date, scheduling, order, completion-time, job, resource, activity, tardiness, machine, due-date, precedence		table constraint, disjunctive, bin-packing, cumulative		OZ, Ilog Scheduler, Cplex	robot, automotive	automotive industry	random instance, real-life, generated instance, industrial partner, benchmark	not-last, energetic reasoning, not-first, edge-finding	1133	1663
LombardiM10a 402	30	due-date, distributed, order, job, make-span, release-date, re-scheduling, task, completion-time, resource, activity, precedence, preempt, scheduling, machine	TCSP	cycle, span constraint, cumulative, disjunctive, table constraint	C	Cplex			real-world, benchmark, real-life	sweep	1160	1690

Manually Extracted Article Features



Table 4: Manually Defined PAPER Properties

Key	Title (Local Copy)	CP System	Bench	Links	Data Avail	Sol Avail	Code Avail	Related To	Classification	Constraints	a	b
AalasnPG23 AalasnPG23 [1] Btt-Monnot23 Btt-Monnot23 [22]	Optimization of Short-Term Underground Mine Planning Using Constraint Programming Enhancing Hybrid CP-SAT Search for Disjunctive Scheduling	CP Opt ARIES CP Opt OR-Tools	real-world real-world, github, benchmark	1	n	y	y	-	JSSP OSSP	-	1	325
EfthymiouY23 EfthymiouY23 [194]	Predicting the Optimal Period for Cyclic Hoist Scheduling Problems	OR-Tools	benchmark, random instance, generated instance, real-life, industrial instance	3	n	n	n	-	CHSP	-	3	415
JuvinHHL23 JuvinHHL23 [528]	An Efficient Constraint Programming Approach to Preemptive Job Shop Scheduling	CP Opt Mistral	supplementary material, github, benchmark	6	ref	y	y	-	PJSSP	endBeforeStart span noOverlap	4	476
JuvinHL23 JuvinHL23 [330]	Constraint Programming for the Robust Two-Machine Flow-Shop Scheduling Problem with Budgeted Uncertainty	CP Opt Cplex	real-world	0	ref	n	-	-	Perm FSSP	endBeforeStart noOverlap sameSequence cumulative	5	477
KameugneFND23 KameugneFND23 [359]	Horizontally Elastic Edge Finder Rule for Cumulative Constraint Based on Slack and Density	?	benchmark	5	BL PSPlib	n	-	-	RCPSPs	-	6	480
KimCMLP23 KimCMLP23 [345]	Iterated Greedy Constraint Programming for Scheduling Steelmaking Continuous Casting	Gurobi OR-Tools	real-world, benchmark, zenodo	0	y	n	-	-	SCC	alternative noOverlap	7	485
Mohdrazab-Somarin23 Mohdrazab-Somarin23 [409]	A Constraint Programming Model for a Reconfigurable Job Shop Scheduling Problem with Machine Availability	CP Opt	random instance	0	n	n	-	-	JSSP RMS	alternative endBeforeStart noOverlap	8	529
PerezaGS123 PerezaGS123 [104]	A Constraint Programming Model for Scheduling the Unloading of Trains in Ports	custom	real-world, generated instance	0	n	n	-	-	SUTP	table disjunctive	9	553
PovedaAA23 PovedaAA23 [506]	Partially Preemptive Multi-Skill/Mode Resource-Constrained Project Scheduling with Generalized Precedence Relations and Calendar	CP Opt MiniZinc Chuffed	real-world, github, benchmark, industrial instance, real-life	4	y	y	-	-	PP-MS- MMRCPSP/max- cal	-	10	557
SquillaciPR23 SquillaciPR23 [564]	Scheduling Complex Observation Requests for a Constellation of Satellites: Large Neighborhood Search Approaches	Cplex Studio	github, benchmark	2	y	n	-	-	EOSP	?	11	584
TardivoDFMP23 TardivoDFMP23 [575]	Constraint Propagation on GPU: A Case Study for the Cumulative Constraint	MiniCPP MiniZinc	bitbucket, github, benchmark, real-world	9	PSPLib BL Pack	y	-	-	RCPSP	cumulative	12	590
TasseelGS23 TasseelGS23 [576]	An End-to-End Reinforcement Learning Approach for Job-Shop Scheduling Problems Based on Constraint Programming	custom Choco	industrial instance, real-world, supplementary material, github, benchmark	0	ref	y	-	-	JSSP	noOverlap	13	591
WangB23 WangB23 [620]	Dynamic All-Different and Maximal Cliques Constraints for Fixed Job Scheduling	FaCiLe	real-world, random instance	0	(y)	n	628	-	FJS	-	14	620
YuraszockMC23 YuraszockMC23 [640]	A competitive constraint programming approach for the group shop scheduling problem	CP Opt	github, benchmark	0	ref	n	-	-	GSSP	noOverlap endBeforeStart	15	633

Extracted Features: Application Areas



Table 16: Works for Concepts of Type ApplicationAreas

Type	Keyword	High	Medium	Low
ApplicationAreas	COVID	GuoZ23 [268]	GebingerKKMMW21 [234]	Fatemi-AnarakiTFV23 [212], Mehdizadeh-Somari23 [430], GurPAE23 [204], JuviniH123a [331], DujanaAYB23 [452], Lemos21 [331]
ApplicationAreas	HVAC	LimHTB16 [390], LimBTB15 [391], GrimesOS14 [266]		
ApplicationAreas	agriculture			AkramNHRS23 [13], BenderWS21 [54], HamPK21 [275], Astrand21 [35], QinWLS21 [311], Astrand21 [26], MeliaY20 [431], PrataAN23 [509], PovedaAA23 [496], Adelgren2023 [7], ErminianesahanGNMS22 [262], Jai-Oh22 [105], ZarandiASC20 [654], HaiderBRPA20 [283], als-1902-69344 [292], Hooker19 [312], LaborierSV18 [374], HookerH17 [312], IranAB16 [334], Lombardi13 [205], Laborier09 [370], KovacsB08 [355], KrogLPH07 [608], MartinPY01 [474], SimonsCK00 [566], GruianK98 [264], Darby-DowmanLMZ07 [163], Wallace06 [645], Simons95 [537], Simons93 [561], PovedaAA23 [506], NaderiRR23 [469], CzerniachowskaW23 [159], NaderiBZ22 [452], NaderiBZ22a [452], AntuoriHHEN21 [22], RobnerGSV21 [313], AkronAPNM21 [164], KochlerBFFHPSS21 [348], VIEHT21 [628], BarzegaranZP20 [61], GebingerMM19 [236], als-1911-04766 [295], BonfiettiZLM16 [113], Sisala5a [552], SchneHH15 [633], AlestoNBG14 [181], HarjunkoskiMBC14 [279], HeinaiBGM06 [58], KovacsV06 [369], Wallace06 [628]
ApplicationAreas	aircraft	PekIAK22 [609], WangB20 [628], TranDRTWOB16 [696], Fahimi16 [205], BajestaniB13 [42], LombardiM12 [405], BajestaniB11 [41], FrankK05 [219], ArtouchineB05 [33], Simons99 [554]	WangB22 [629], GombolayWS18 [253], Ham18 [273], Simons07 [559], SakkottW00 [529], Simons95a [556]	
ApplicationAreas	automotive		GuoZ23 [268], YuraszeckMPV22 [650], EmdeZD22 [109], Groleaz21 [261], LimtanyakulS12 [303], SomLYL10 [567], Lombardi10 [335], BarlatCG08 [52], SchildW00 [532]	
ApplicationAreas	cable tree	KochlerBFFHPSS21 [348]		
ApplicationAreas	car manufacturing		AntuoriHHEN21 [22]	
ApplicationAreas	container terminal	QinDCS20 [512], SacramentoSP20 [526]	LatorreRSV18 [372]	BeldiceanuC94 [78], als-2312-13638 [421], PetrovCSL23 [496], TomaiB122 [592], CauwelactDS20 [142], WallaceY20 [627], ZarandiASC20 [654], FallahiAC20 [289], Hooker19 [312], CauwelactDMS16 [149], DeJemeppe16 [174], DeJemeppeCS15 [173], NorasH12 [476], CorreaL07 [153], LimTK04 [339], NaderiRR23 [469], WangB22 [629], Adelgren2023 [7], ErminianesahanGNMS22 [262], NaderiBZ22a [456], NaderiBZ22 [452], HeinaiNV12 [605], Jai-Oh22 [105], Leino21 [351], MokhtarianZV19 [29], Jai-Oh22 [105], TranJWSK18 [574], HookerH17 [312], DoughabRP16 [120], LipovetkyBPS14 [304], HachemiGR11 [274], MilanoW09 [441], WuBB09 [843], MilanoW06 [440], BeldiceanuC02 [79], JainG01 [523], SimouletK00 [260], Bartak02 [54], Bartak02a [53], Groleaz21 [261], Zahouti21 [652], GallegherKSBB10 [225], Medi-WambolOHM17 [418], Latorre13 [382], IfrimOS12 [339], LatorreBC12 [383]
ApplicationAreas	crew-scheduling	ZarandiASC20 [654], PourDEB18 [509]	BourreauGGL122 [113], Zahouti21 [652], GombolayWS18 [253], Masam01 [429], Touraivane03 [593]	
ApplicationAreas	dairies			
ApplicationAreas	dairy	EscobetPQIRA19 [201]	PrataAN23 [509], HarjunkoskiMBC14 [279]	
ApplicationAreas	datacenter	HermesierJL11 [306]		
ApplicationAreas	datacentre		HurleyOS16 [319]	
ApplicationAreas	day-ahead market			
ApplicationAreas	deep space			
ApplicationAreas	drone	MontemanniD23a [446], MontemanniD23 [447], Ham18 [273]		HebrardALLOMR22 [285], GuoZ23 [269], JuviniH123a [331], Adelgren2023 [7], ShahKH23 [547], EmdeZD22 [109], Astrand21 [35], Astrand21 [36], AntuoriHHEN21 [22], ZarandiASC20 [654], Ham18a [274]

Table 8: Co-Authors of Articles/Papers

Author	Nr Works	Nr Cites	Entries
J. Christopher Beck	49	701	LuoB22 416 , ZhangBH22 658 , TangB20 573 , RoshanetBAUB20 521 , TranPZLDB18 597 , TranVNB17 599 , TranVNB17a 600 , CohenHB17 154 , BoothNB16 114 , KOB16 865 , TranAB16 594 , TranWDRFOVP16 601 , LuoVLBDM16 415 , TranDRWVOVB16 596 , BajostanB15 43 , KoschB14 835 , TerekhovDB14 581 , LouieNB14 412 , HeinzSB13 294 , HeinzKB13 291 , BajostanB13 42 , TranDB13 598 , HeinzB12 290 , TerekhovDOB12 580 , TranB12 595 , ZarandB12 213 , KowasB11 289 , HeckmanB11 289 , BajostanB11 41 , WuB109 543 , RiderVLB09 74 , CaracheaB09 731 , WatsonB08 659 , KowasB08 659 , BazAW07 74 , Beck07 654 , KowasB07 654 , Beck06 652 , CaracheaB05 132 , WuB05 642 , BeckW05 72 , BeckW04 71 , BeckR03 130 , BeckPS03 69 , BeckP06 65 , BeckP09 62 , BeckP08 67 , BeckD97 65
Michela Milano	31	297	BorghesiBLMB18 115 , BonfiettiZLM16 113 , BridiBLMB16 120 , BridiLBBM16 121 , LombardiBM15 399 , BartoliniBLM14 60 , BonfiettiLM14 111 , BonfiettiLBM14 109 , BonfiettiLM13 406 , LombardiMB13 407 , LombardiB12 405 , BonfiettiB12 108 , LombardiM12a 403 , BonfiettiM12 112 , BonfiettiLBM11 107 , LombardiMBM11 100 , BeniniLMR11 99 , MilanoI 458 , LombardiM10 403 , LombardiM10a 402 , LombardiMRB10 408 , LombardiM09 401 , RuggieroBBM09 525 , MilanoW09 441 , BeniniLMR08 89 , BeniniBGM06 88 , MilanoW06 140 , MilanoORT02 439 , LammaMM07 374 , BrusoniCLMM196 124
Andreas Schutt	27	322	WuSS19 644 , KreterSS18 364 , GoldwasserSS18 251 , MusliuSS18 454 , KreterSS17 363 , YounisFS17 649 , GoldwasserS17 250 , SchuttB16 643 , SerediS16 670 , KreterSS15 362 , EvensSH15 203 , EvensSH15a 204 , SchuttFSW15 642 , ThiruvadyWGS14 585 , GuSWC14 266 , SchuttFS13 537 , SchuttPS13a 536 , GuSS13 265 , SchuttFSW13 641 , ChuGNSW13 147 , SchuttCSW12 535 , SchuttFSW11 640 , SchuttB11 534
Michele Lombardi	25	194	BorghesiBLMB18 115 , CauwelaertLS18 111 , BonfiettiZLM16 113 , BridiBLMB16 120 , BridiLBBM16 121 , LombardiBM15 399 , BartoliniBBLM14 60 , BonfiettiLM14 111 , BonfiettiLBM14 109 , BonfiettiLM13 406 , LombardiMB13 407 , LombardiM12 405 , BonfiettiLBM12 108 , LombardiM12a 403 , BonfiettiM12 112 , BonfiettiLBM11 107 , LombardiMBM11 100 , BeniniLMR11 99 , LombardiM10 403 , LombardiM10a 402 , LombardiM10 998 , LombardiMRB10 408 , LombardiM09 401 , BeniniLMR08 89 , HoeveGSL07 609
Peter J. Stuckey	24	453	WuSS19 644 , DemirovicSS18 117 , KreterSS18 364 , MusliuSS18 455 , KreterSS17 363 , SchuttB16 643 , BlomPS15 100 , KreterSS15 362 , BurtLPS15 124 , SchuttFSW15 642 , BlomBPS14 99 , GuSWC14 266 , SchuttFS13 537 , SchuttPS13a 536 , SchuttGS13 265 , SchuttFSW13 641 , SchuttCSW12 535 , GuSW12 267 , SchuttFSW11 640 , BandaSC11 170 , BandaSC11 109 , BandaSC11 109-0347 539 , SchuttFSW09 538 , OhrimenkoSC09 483
John N. Hooker	19	1316	EckO022 1057 , HookerH19 872 , HookerH17 874 , HookerH16 288 , CirechiH16 150 , HarjunenMBC14 278 , CirechiH14 149 , CobanH11 158 , CobanH10 154 , HookerH10 310 , HookerH9 309 , HookerH8 308 , HookerH7 306 , HookerH6a 307 , HookerH4 305 , HookerO03 313 , HookerY02 313 , HookerO01 303
Emmanuel Hebrard	17	71	JuviniHLH23 328 , HebrardALLCMR22 285 , AntuoriHHEN21 22 , ArtiguesHQT21 52 , GodelLH20 247 , AntuoriHHEN20 21 , HebrardHJMPV16 280 , SimoninAHL15 553 , GrimesH15 258 , BessiereHMQW14 53 , SimoninAHL12 554 , BillautHL12 554 , GrimesH11 257 , GrimesH10 256 , GrimesH09 259 , HebrardW05 257
Pierre Lopez	17	90	JuviniHLH23 328 , JuviniHLH22 329 , HebrardALLCMR22 285 , JuviniHLH22 329 , Polo-MejiaALB20 503 , NattaHKA19 469 , NattaAL17 463 , NattaALR16 464 , SimoninAHL15 555 , NattaAL15 462 , PolaBHL15 462 , BillautHL11 375 , TrojetHL11 662 , LopezAKYG08 410 , TorresL00 591
Christian Artigues	16	203	PovedaA23 500 , PolaAK22 502 , HebrardALLCMR22 285 , ArtiguesHQT21 52 , Polo-MejiaALB20 503 , NattaHKA19 469 , NattaAL17 463 , NattaALR16 464 , SimoninAHL15 554 , NattaAL15 462 , SialaAHL15 554 , SimoninAHL12 554 , NeronABCD06 481 , DemassyA05 459 , ArtiguesH10 41 , ArtiguesH09 40
Pierre Schaus	15	669	CauwelaertDS20 114 , ThomasKS20 589 , HoudjSW19 316 , CappartTSR18 1130 , CauwelaertLS18 1141 , CappartS17 129 , CauwelaertDMS16 140 , DejeneppeCS15 1131 , GayHLS15 229 , GayHLS15 229 , GayHLS15a 231 , HoudjSWD14 317 , GayS14 232 , SchausHM1 631 , SchausD08 530
Helmut Simonis	15	154	ArmstrongCOS22 271 , ArmstrongGOS21 60 , AntunesABD20 28 , AntunesABD18 19 , HurleyOS16 919 , SimonisOS14 268 , IfrimOS12 528 , SimonH11 522 , SimonP7 522 , SimonP7 522 , SimonP09 523 , SimonP05 521 , SimonP05a 522 , DinculescuP04 154
Nicolas Beldiceanu	13	274	Madi-WambsLBM17 115 , Madi-WambsLBM16 117 , LefortCB15 385 , LefortCB15 384 , LefortCB12 383 , CiereqPBB11 151 , BeldiceanuCDP11 80 , BeldiceanuCP08 108 , MadiP08 600 , BeldiceanuP07 82 , PodersB04 501 , BeldiceanuC02 79 , AgoumB93 9
Luca Bentini	13	146	BorghesiBLMB18 115 , BridiBLMB16 120 , BridiLBBM16 121 , BonfiettiLBM14 109 , LombardiMB13 407 , LombardiB12 405 , BonfiettiB12 108 , BonfiettiLBM11 107 , LombardiMBM11 100 , BeniniLMR11 99 , LombardiMRB10 408 , RuggieroBBM09 524 , BeniniLMR08 89 , BeniniBGM06 88 , LamariniL04 29 , LatorreCS18 172 , LatorreCS18 172 , LatorreCS18 171 , VilimBC05 620 , LatorreCS07 370 , BilodevLB06 94 , BaptisteLPN06 47 , NeronABCD06 481 , GodardN05 243 , LatorreCS07 369 , FocacciN00 215
Philippe Laborie	12	513	LamariniL04 29 , LatorreCS18 172 , LatorreCS18 172 , LatorreCS18 171 , VilimBC05 620 , LatorreCS07 370 , BilodevLB06 94 , BaptisteLPN06 47 , NeronABCD06 481 , GodardN05 243 , LatorreCS07 369 , FocacciN00 215
Philippe Baptiste	11	403	BaptisteB18 49 , Baptiste09 45 , BaptisteLPN06 47 , NeronABCD06 481 , ArtiguesH05 34 , Baptiste02 44 , BaptisteP01 50 , BaptisteP00 49 , PapaB08 492 , BaptisteP97 48 , PapaB97 491
Roman Barták	11	88	SvancaraB22 599 , JelinekHB16 325 , BartakV15 59 , BartakI4 55 , BartakS11 57 , BartakCS10 56 , BartakSR10 58 , VilimBC05 620 , VilimBC04 619 , Bartak02 64 , Bartak02a 65



- Limited coverage by OpenCitations
- Difficult to have local access to some publication types (book, incollection)
- Heavily biased towards publications in English
- More powerful NLP analysis of works possible?

Problem: Count for Most Cited Papers



Table 9: Works from bibtex (Total 30)

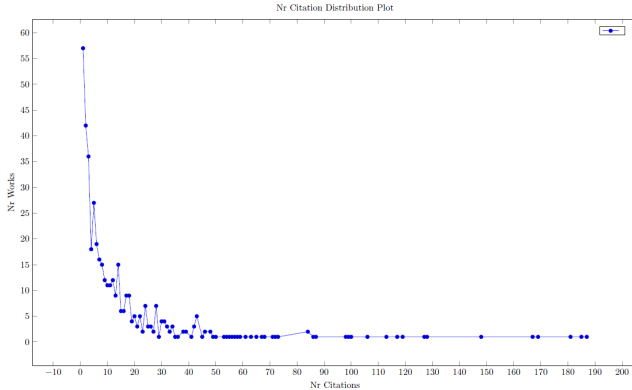
Key	Authors	Title	LC	Cite	Year	Conference / Journal	Pages	Nr Cites	Nr Refs	b	c
JainM99 JainM99	A. Jain, S. Mooran	Deterministic job-shop scheduling: Past, present and future	Yes	[322]	1999	European Journal of Operational Research	45	490	150	1352	1753
HarjunksoskiMBC14	I. Harjunksoski, Christos T. Maravelias, P. Boazerg, Pedro M. Castro, K. Engell, Ignacio E. Grossmann, John N. Hooker, C. McGolden, G. Sauli, J. Wasicki	Scope for industrial applications of production scheduling models and solution methods	Yes	[279]	2014	Computers Chemical Engineering	33	381	176	1335	1649
BlazewiczDP96	J. Blazewicz, W. Domschke, E. Pesch	The job shop scheduling problem: Conventional and new solution techniques	Yes	[125]	1996	European Journal of Operational Research	33	344	127	1278	1762
HookerO03 HookerO03	John N. Hooker, G. Ottosson	Logic-based Benders decomposition	Yes	[313]	2003	Mathematical Programming Book	28	317	0	1347	1729
BaptistePN01	P. Baptiste, Claude Le Pape, W. Nuijten	Constraint-Based Scheduling	No	[50]	2001		null	296	0	No	n/a
JainG01 JainG01	V. Jain, Ignacio E. Grossmann	Algorithms for Hybrid MILP/CP Models for a Class of Optimization Problems	Yes	[323]	2001	INFORMS Journal on Computing	19	279	23	1351	1738
AggounB93 AggounB93	A. Aggoun, N. Beldiceanu	Extending CHIP in order to solve complex scheduling and placement problems	Yes	[2]	1993	Mathematical and Computer Modelling	17	187	11	1247	1767
Hooker00 Hooker00	John N. Hooker	Logic Based Methods for Optimization: Combining Optimization and Constraint Satisfaction	No	[304]	2000	Book	null	185	0	No	n/a
Hooker07 Hooker07	John N. Hooker	Planning and Scheduling by Logic-Based Benders Decomposition	Yes	[309]	2007	Operations Research	29	181	19	1345	1715
HarjunksoskiG02	I. Harjunksoski, Ignacio E. Grossmann	Decomposition techniques for multistage scheduling problems using mixed-integer and constraint programming methods	Yes	[275]	2002	Computers Chemical Engineering	20	169	11	1334	1733
BeldiceanuC94	N. Beldiceanu, E. Contejean	Introducing Global Constraints in CHIP	Yes	[75]	1994	Mathematical and Computer Modelling	27	167	8	1271	1765
LaborieRSV18	P. Laborie, J. Rogerio, P. Shaw, P. Vilim	IBM ILOG CP optimizer for scheduling - 20+ years of scheduling with constraints at IBM/ILOG	Yes	[372]	2018	Constraints An Int. J.	41	148	35	1370	1610
Laborie03 Laborie03	P. Laborie	Algorithms for propagating resource constraints in AI planning and scheduling: Existing approaches and new results	Yes	[369]	2003	Artificial Intelligence	38	128	10	1369	1731
OhrimenkoSC09	O. Ohrimenko, Peter J. Stuckey, M. Codish	Propagation via lazy clause generation	Yes	[483]	2009	Constraints An Int. J.	35	127	15	1417	1702
KuB16 KuB16	W. Ku, J. Christopher Beck	Mixed Integer Programming models for job shop scheduling: A computational analysis	Yes	[365]	2016	Computers Operations Research	9	119	17	1367	1630
Rodriguez07 Rodriguez07	J. Rodriguez	A constraint programming model for real-time train scheduling at junctions	Yes	[320]	2007	Transportation Research Part B: Methodological	15	117	6	1430	1716
LiW08 LiW08	H. Li, K. Womer	Scheduling projects with multi-skilled personnel by a hybrid MILP/CP benders decomposition algorithm	Yes	[386]	2008	Journal of Scheduling	18	113	31	1374	1708
CorreaLR07	Ayoub Insa Corréa, A. Langevin, L. Rousseau	Scheduling and routing of automated guided vehicles: A hybrid approach	Yes	[158]	2007	Computers Operations Research	20	106	20	1296	1714
MengZRLZ20	L. Meng, C. Zhang, Y. Ren, B. Zhang, C. Lv	Mixed-integer linear programming and constraint programming formulations for solving distributed flexible job shop scheduling problem	Yes	[433]	2020	Computers Industrial Engineering	13	100	62	1393	1574
BensanaIV99	E. Bensana, M. Lemaître, G. Verfallide	Earth Observation Satellite Management	Yes	[61]	1999	Constraints An Int. J.	7	99	0	1276	1752

OpenCitation Count Compared to Google Scholar



Key	Type	Google	OC	Ratio
JainM99	article	1116	490	2.28
HarjunkoskiMBC14	article	588	381	1.54
BlazewiczDP96	article	796	344	2.31
BaptistePN01	book	1039	296	3.51
AggounB93	article	502	187	2.68
LaborieRSV18	article	309	148	2.09
BensanaLV99	article	251	99	2.54
DincbasSH90	article	271	86	3.15
Thorsteinsson01	paper	205	67	3.06
DincbasSH88	paper	287	0	☹

Problem: Citation Count Distribution



Outline



CP and Scheduling Literature Survey

Summary



- Use the survey to find
 - Most important works on Constraint Based Scheduling
 - Specialized papers on the constraint reasoning for scheduling
 - Works in specific application domains or specific industries